

Expanding Double Brackets

AQA KS3 · Year 8 · Foundation

To expand double brackets, multiply every term in the first bracket by every term in the second bracket. You can use the FOIL method (First, Outside, Inside, Last) or the grid method to organise your working. Then collect like terms to simplify.

Method

- 1 Multiply each term in the first bracket by each term in the second
- 2 Write each product in its grid cell
- 3 Collect like terms (the two middle terms)
- 4 Write the simplified answer
- 5 Check by substituting a value

Common mistakes

- Forgetting the last term: you must multiply the constant in the first bracket by the constant in the second.

Worked example

Expand and simplify: $(x + 2)(x + 3)$

Step 1 — Draw the grid with $(x + 2)$ down the side and $(x + 3)$ along the top.

Step 2 — Fill in: $x \cdot x = x^2$, $x \cdot 3 =$

$3x$, $2 \cdot x = 2x$, $2 \cdot 3 = 6$

Visual aid — grid method

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$3x2x6$

Grid model showing $(x+2)$
 $(x+3)$: top row x times x
gives x squared, x times
 $+3$ gives $3x$; bottom row $+2$
times x gives $2x$, $+2$ times

Questions

Expand and simplify — show your working in your workbook

Q1 of 8

[1] ● ○ ○ ○ ○

Expand and simplify: $(x + 1)(x + 2)$

Q2 of 8

[1] ● ○ ○ ○ ○

Expand and simplify: $(x + 3)(x + 4)$

Q3 of 8

[2] ● ● ○ ○ ○

Expand and simplify: $(x + 5)(x - 2)$

Q4 of 8

[2] ● ● ○ ○ ○

Expand and simplify: $(x - 3)(x + 4)$

Q5 of 8

[2] ● ● ● ○ ○

Expand and simplify: $(x - 2)(x - 5)$

Q6 of 8

[3] ● ● ● ○ ○

Expand and simplify: $(2x + 1)(x + 3)$

Q7 of 8

[3] ● ● ● ● ○

Expand and simplify: $(3x - 2)(x + 4)$

Q8 of 8

[3] ● ● ● ● ●

Expand and simplify: $(2x + 3)(3x - 1)$